Table 1 | Characteristics of included studies

Author (Year)	Country	Clinical Domain	Clinical Task	Study Design	Testing (n)	AI Architecture	Explanation Modality	Delivery Format	Patient Involvement Phase
Alam & Mueller (2021)	USA	Internal Medicine	Diagnosis	Randomised Experiment	193	Simulated Algorithm	Hybrid	Web Platform	Summative Testing
Alduhailan et al. (2025)	Saudi Arabia	Primary & Emergency Care	Triage	Observational Study	35	Black-Box ML	Textual	Mobile Health App	Summative Testing
Algarni (2025)	Saudi Arabia	Radiology	Diagnosis	Observational Study	Not Reported	Large Language Model	Hybrid	Conceptual Prototype	Summative Testing
Aziz et al. (2026)	Pakistan	Psychiatry	Self-Management	Randomised Controlled Trial	120	Large Language Model	Textual	Web Platform	Clinical Implementation
Begum et al. (2025)	Pakistan	Internal Medicine	Self-Management	Observational Study	200 (20)*	Black-Box ML	Textual	Mobile Health App	Clinical Implementation
Bolpagni et al. (2024)	Italy	Psychiatry	Self-Management	Qualitative Study	12	Simulated Algorithm	Textual	Conceptual Prototype	Co-design
Gombolay et al. (2024)	USA	Internal Medicine	Treatment Selection	Randomised Experiment	365	Black-Box ML	Visual	Web Platform	Summative Testing
Gould et al. (2023)	Australia	Surgery	Prognosis	Qualitative Study	20	Simulated Algorithm	Textual	Conceptual Prototype	Needs Assessment
He et al. (2023)	China	Internal Medicine	Screening	Mixed Methods	8	Simulated Algorithm	Hybrid	Conceptual Prototype	Co-design
Hildebrand et al. (2023)	USA	Oncology	Prognosis	Qualitative Study	30	Black-Box ML	Textual	Static Report	Summative Testing
Huang et al. (2024)	USA	Primary & Emergency Care	Self-Management	Randomised Experiment	155	Large Language Model	Textual	Web Platform	Summative Testing
Lombaers et al. (2026)	Netherlands	Oncology	Prognosis	Qualitative Study	19	Interpretable Machine Learning	Hybrid	Static Report	Needs Assessment
Metta et al. (2024)	Italy & France	Oncology	Diagnosis	Observational Study	156	Black-Box ML	Hybrid	Web Platform	Summative Testing
Sperling et al. (2025)	USA	Internal Medicine	Prognosis	Qualitative Study	52	Simulated Algorithm	Textual	Conceptual Prototype	Needs Assessment
Tsai et al. (2021)	USA	Primary & Emergency Care	Triage	Mixed Methods	45	Interpretable Machine Learning	Textual	Web Platform	Needs Assessment
Wang et al. (2025)	China	Psychiatry	Screening	Randomised Experiment	346	Simulated Algorithm	Textual	Web Platform	Summative Testing
Williams et al. (2025)	USA	Psychiatry	Screening	Qualitative Study	33	Interpretable Machine Learning	Hybrid	Conceptual Prototype	Co-design
Woodcock et al. (2021)	United Kingdom	Primary & Emergency Care	Triage	Randomised Experiment	750	Simulated Algorithm	Textual	Web Platform	Summative Testing
Wu et al. (2025)	China	Surgery	Screening	Randomised Controlled Trial	70	Large Language Model	Hybrid	In-Clinic App	Summative Testing
Zhang et al. (2021)	USA	Radiology	Self-Management	Randomised Experiment	3,423	Interpretable Machine Learning	Textual	Web Platform	Summative Testing

** Testing N reflects longitudinal cohort (N=200); sub-sample (n=20) completed usability testing.*